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Oefeningenreeks 3C nr. 17.7

$$\lim_{x\to 0} \frac{x\sin^2 x}{x\cos x - \sin x}$$
$$= \frac{0}{0} \Rightarrow \text{l'Hôpital}$$
$$= \lim_{x\to 0} \frac{\sin^2 x + 2x\sin x\cos x}{\cos x - x\sin x - \cos x}$$
$$= \lim_{x\to 0} \frac{\sin x\sin x + 2x\sin x\cos x}{-x\sin x}$$
$$= \lim_{x\to 0} \frac{\sin x + 2x\cos x}{-x}$$
$$= -\lim_{x\to 0} \left(\frac{\sin x}{x} + 2\cos x\right)$$
$$= -(1+2) = -3$$

References